



DPUSGEWeather

Displays weather effects such as rain and thunderstorms within the visualization SGE.

1. Inputs:

<i>ExternalTimeStamp</i>	Input used for synchronization. If <i>DPUSGEWeather</i> runs on more than one graphics computer in parallel, it should be connected to a uniform time source within the whole simulator (e.g. the output <i>Trafficdata</i> of <i>DPUTRFX</i>).
<i>v</i>	Current speed of the EGO vehicle (m/s).

2. Dynamic parameters:

<i>Rain.Speed</i>	Rain fall speed [m/s] (Default: 20)
<i>Rain.Length</i>	Length of a raindrop [m] (Default: 0.7)
<i>Rain.Height</i>	Height from which the rain falls [m] (Default: 20)
<i>Rain.ViewAngle</i>	Half field of view [°] (Default: 30)
<i>Rain.ViewSize</i>	Maximum distance for raindrops [m] (Default: 50)
<i>Rain.Color</i>	Grey value of the color (Default: 0.88, range: 0..1)
<i>Rain.Alpha</i>	Rain transparency (Default: 0.5)
<i>Rain.MaxDensity</i>	Maximum number of raindrops (Default: 20000)
<i>Rain.MinDensityMul</i>	Factor for maximum density when the vehicle is not moving (Default: 0.5)
<i>Rain.VolMax</i>	Maximum sound volume (Default: 1.0)
<i>Rain.VolMul</i>	Multiplier for rain sound volume (Default: 1.0)
<i>Thunder.VolMin</i>	Minimum volume for distant thunder (Default: 0.2)
<i>Thunder.VolMax</i>	Maximum volume for close thunder (Default: 1.0)
<i>Thunder.DelayMin</i>	Delay for close thunder, relative to the flash [s] (Default: 0.5)
<i>Thunder.DelayMax</i>	Delay for distant thunder [s] (Default: 9)
<i>Flash.OpacityMax</i>	Maximum brightness of the flash (Default: 0.3)
<i>Flash.OpacityMin</i>	Minimum brightness of the flash (Default: 0.03)
<i>Flash.MaxFrequency</i>	Frequency of flashes at maximum thunderstorm strength [1/s] (Default: 0.1)
<i>Flash.DurationMS</i>	Duration of flash display [ms] (Default: 100)

3. Parameters:

<i>Precipitation</i>	Amount of precipitation (rain) (0..1)
<i>Thunderstorm</i>	Strength of thunderstorm (flash and thunder) (0..1)



There are many additional DPU parameters for the weather simulation's individual components, which can be used to tailor the display. As the parameters have meaningful default values, this is however usually not necessary.

4. Outputs:

<i>Rain.Volume</i>	Current rain sound volume. Can be connected to an instance of <i>DPUSNDXWavPlayer</i> .
<i>Thunder.Volume</i>	Current thunder sound volume. Can be connected to an instance of <i>DPUSNDXWavPlayer</i> .
<i>Flash</i>	Boolean flag: has a flash been generated during this simulation step?
<i>Thunder</i>	Boolean flag: has thunder been generated during this simulation step?

The weather simulation is mainly controlled using parameters of the SILAB environment control (see manual for *DPUEnvControl*).

5. Dependencies to other DPUs:

To run **DPUSGEWeather**, the DPUs **DPUSGEWorld** and **DPUSGEView** are additionally required. The Index value (computation sequence) of **DPUSGEWeather** must be higher than **DPUSGEWorld**'s, but lower than **DPUSGEView**'s.

6. Example:

The following script is a minimal example with which **DPUSGEWeather** can be used:

```
DPUSGEWeather SGEWeather
{
    Computer = {LOCALHOST};
    Index = 55;

    Rain.ViewAngle = 40;

    Rain.MaxDensity = 20000;
    Rain.VolMax = 0.3;
    Rain.Speed = 20;
    Rain.Length = 0.7;
};

Connections =
{
    VDyn.v -> SGEWeather.v,
    TRFX.Trafficdata -> SGEWeather.ExternalTimeStamp
};
```

This script sets the parameters for the rain inside the simulation.



As noted before, the velocity $VDyn.v$ of the vehicle is connected to our DPU's input v . The same holds for *ExternalTimeStamp*, to which a uniform time source within the whole simulator, namely *TRFX.Trafficdata*, is connected.